

EMS Data Analysis and Forecasting

For CSIS 4495 Applied Research, Section 002

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Introduction

The Emergency Medical Services (EMS) play a critical role in the healthcare system by providing rapid, pre-hospital care in unpredictable and often hazardous environments (Reichard et al., 2017; Maguire et al., 2023). EMS clinicians routinely lift and move patients, work in cramped or unsafe scenes, and face time pressure and environmental stressors. These conditions expose them to a wide range of occupational hazards, including musculoskeletal strain, blood and body-fluid exposures, violence, and transportation incidents (Reichard et al., 2017; Marsh et al., 2025). Multiple studies have shown that EMS workers experience substantially higher rates of work-related injuries and exposures than the general workforce and even other healthcare workers (Reichard et al., 2017; Maguire et al., 2023; Xia & Collie, 2020).

At the same time, EMS systems must manage fluctuating call volumes, seasonal patterns, and special events that create workload peaks. When staffing and resources are not aligned with these peaks, individual clinicians can experience high cognitive and physical load, fatigue, and rushed task performance. Prior research in both EMS and hospital settings suggests that higher workload and fatigue are associated with increased risk of injuries, musculoskeletal disorders, and safety incidents (Alhazmi et al., 2024; Hammad et al., 2024; Reichard et al., 2017). However, most EMS agencies still monitor injury rates retrospectively and rely on simple descriptive statistics or lagging indicators. They often discover “bad weeks” only after a cluster of exposure or injury reports has already occurred (Reichard et al., 2017; National EMS Information System [NEMSIS], 2025).

This project focuses on one component of this broader safety challenge: predicting when work-related exposure events are likely to increase, using routinely collected EMS data. The central problem can be framed as follows:

- Can we build a time-series model that forecasts weekly counts of work-related exposures among EMS providers over the next few weeks?
- Are there temporal patterns (e.g., trends or seasonality) in exposure events that can inform proactive staffing and safety planning (Qian et al., 2016; Tarisayi et al., 2024)?
- How can these forecasts be presented in a simple product that EMS managers can interpret and act on?

Answering these questions is important because a reliable forecast of exposure risk— even a simple one—could enable EMS leaders to plan staffing, training, and safety interventions before high-risk periods, rather than reacting after injuries occur.

Background and Related Work

Several strands of literature motivate this work. Occupational health studies have documented high rates of injuries and exposures among EMS clinicians, often linked to overexertion, awkward lifting, transportation incidents, and violence (Reichard et al., 2017; Maguire et al., 2023; Marsh et al., 2025). Recent surveys indicate that more than a quarter of EMS clinicians report an occupational injury and more than a third report an occupational exposure within a 12-month period, with risk modified by agency characteristics and mitigation strategies such as lifting policies and ergonomic equipment (Alhazmi et al., 2024; Hammad et al., 2024; Maguire et al., 2023). Other studies have highlighted the prevalence of work-related musculoskeletal disorders and their association with physically demanding tasks such as lifting, carrying, and awkward postures (Marsh et al., 2025; Alhazmi et al., 2024).

In parallel, research in emergency departments and other acute-care settings has explored forecasting workload using time-series models. For example, SARIMAX models have been used to predict emergency department occupancy several hours ahead using historical counts and operational indicators, outperforming simple rolling averages and providing useful prediction intervals for real-time decision making (Qian et al., 2016; Tarisayi et al., 2024). Similar approaches have been applied to predict patient falls and other adverse events using electronic health record and operational data (Qian et al., 2016; Tarisayi et al., 2024; Vieira et al., 2023). These studies show that time-series forecasting can be integrated into operational dashboards and used to allocate resources more efficiently.

However, there is relatively limited work that links time-series forecasting directly to occupational injury or exposure risk in EMS. Most EMS safety studies are cross-sectional or retrospective, describing prevalence and risk factors but not providing forward-looking tools that operations teams can use day-to-day (Reichard et al., 2017; Maguire et al., 2023). National data sources like NEMSIS contain elements indicating whether an EMS event involved a suspected work-related exposure (eOther.06), but they do not consistently capture detailed injury cause or type at the national level, limiting the ability to model specific mechanisms of harm. This creates a gap between what we know about injury patterns and the practical tools available to prevent them.

Knowledge Gaps and Project Focus

This project aims to address two main gaps:

1. Lack of operational forecasting tools for EMS staff exposures.

Existing research documents that EMS workers are at high risk of exposure and injury, but most agencies lack simple, interpretable tools that forecast when exposure events are likely to rise based on their own data (Reichard et al., 2017; Maguire et al., 2023). By focusing on

weekly counts of work-related exposures, this project seeks to turn retrospective surveillance data into a forward-looking signal.

2. Limited use of national EMS data for proactive safety planning.

National NEMSIS data include a work-related exposure flag that can be aggregated over time, but they do not contain detailed provider injury cause/type at the national level, and operational time fields require substantial preprocessing (NEMSIS, 2019; NEMSIS, 2025). This project explores how far we can go using the elements that are available consistently (e.g., exposure flag plus time stamps) to build a forecasting product, while explicitly acknowledging these limitations.

Because detailed provider injury mechanisms are not available in the national dataset, the project deliberately narrows its scope to forecasting exposure events over time, rather than predicting specific injury types or severities (NEMSIS, 2019; NEMSIS, 2025). The emphasis is on building a product that is robust to data limitations and can still inform safety-oriented decisions.

Initial Hypotheses, Assumptions & Potential Benefits

Based on prior literature, operational experience, and an initial broader hypothesis set, the project originally considered seven hypotheses related to workload, staffing, shift length, fatigue, ergonomics, equipment, and safety culture. Those early ideas were useful for framing how operational factors might influence injury risk, but most require detailed staffing, scheduling, equipment, or culture data that are not available or reliable in the national NEMSIS research dataset. Given the actual data elements that can be used consistently at the national level—particularly the EMS work-related exposure flag and time fields—the project has narrowed its empirical focus to three testable hypotheses centered on temporal patterns, forecastability, and operational usefulness of exposure-risk forecasts.

Hypotheses

ID	Hypothesis	Description
H1	Temporal patterns in exposures	Weekly counts of work-related exposures will exhibit trend and/or seasonal patterns over 2024 (e.g., higher risk during specific months or higher-workload periods).
H2	Forecastability	With limited covariates, time-series models such as SARIMA or Prophet can produce usable short-term forecasts (about 4–12 weeks ahead) of weekly exposure counts, including prediction intervals to reflect uncertainty.
H3	Operational usefulness	Displaying these forecasts in an interactive dashboard—with ranges and “original vs updated” views as new data arrive—will be more actionable for EMS managers than static tables of historical injury rates.

Key Assumptions

Data sufficiency: The approximately 20,000 exposure-flagged events in the 2024 NEMSIS data will be sufficient to construct a stable weekly time series once appropriate time fields are joined and “Not Applicable/Not Recorded” values are filtered out.

Potential Benefits

If these hypotheses hold, the potential benefits include:

- Giving EMS leaders an early warning signal of upcoming high-risk weeks, so they can pre-emptively adjust staffing, training, or safety protocols.
- Demonstrating how national EMS data can be repurposed from retrospective reporting to proactive risk forecasting.
- Providing a flexible product framework: if advanced models are not supported by the data, the same interface can still host simpler forecasting methods or monitoring rules, preserving value for decision makers.

Summary of the Initially Proposed Research

This project uses the National Emergency Medical Services Information System (NEMSIS) Public-Use Research Dataset to build data-driven tools that forecast EMS demand peaks and identify conditions linked to paramedic injuries (NEMSIS, 2024; NEMSIS, 2025). NEMSIS is a large, de-identified national database with tens of millions of EMS activations that supports research on EMS utilization, safety, and system performance (National Highway Traffic Safety Administration, 2025). Motivated by evidence that EMS workers experience high rates of musculoskeletal and other occupational injuries compared with other sectors, the project focuses on how forecasting and automated reporting can be used to support quality assurance (QA) and reduce disabling injuries, defined as injuries causing at least one missed workday (Reichard et al., 2017; Marsh et al., 2025; Defence Research and Development Canada, 2017).

The core problem is framed around three questions: how to leverage NEMSIS data to forecast EMS demand and peak loads; which models and dashboards can uncover injury-linked patterns such as overloads that contribute to accidents; and what QA improvements, supported by AI/ML forecasting and automated reporting, can reduce field injuries (Jones, 2024; Shahidian et al., 2025; Qian et al., 2016). The project emphasizes Canadian EMS context and disabling injury rates, noting that some services report disabling injury rates exceeding 20% annually, underscoring the importance of proactive prevention (Defence Research and Development Canada, 2017; Edmonton Metro EMS, 2025). The literature motivating the proposal includes reports on musculoskeletal disorders from patient handling, high volumes of workers’ compensation claims in paramedic

services, and significant exposure to violence and psychological trauma (Reichard et al., 2017; Yolland, 2001; Defence Research and Development Canada, 2017).

Methodologically, the project adopts a five-phase data science workflow: exploratory data analysis, data cleaning and pipeline development, model building, automated reporting, and dashboard design. The data source is the 2019–2024 NEMSIS Public-Use Research Dataset, which contains over 50 million EMS incidents per year, though the project will work with a stratified sample of about 200,000 incidents to fit within time and compute constraints (NEMSIS, 2024; NEMSIS, 2025). The analysis plan includes outlier detection, missing data handling, feature engineering (e.g., hour-of-day, day-of-week, call type, crew size), and testing several models (linear and logistic regression, random forests, gradient boosting) to forecast demand peaks and identify high-risk periods (Jones, 2024; Shahidian et al., 2025; Vieira et al., 2025).

Technically, the work will rely on Python (Pandas, scikit-learn, XGBoost), Jupyter Notebooks (with possible Databricks exploration), and a Streamlit dashboard front-end (Jones, 2024; Vieira et al., 2025). Expected outputs include cleaned datasets, forecasting models targeting at least 75% accuracy for demand peaks, an automated rule-based QA report generator, and interactive dashboards highlighting busy periods and injury “hot spots” (Jones, 2024; Shahidian et al., 2025). The anticipated practical impact is better crew scheduling, reduced fatigue-related injuries and claims, and improved patient care, as well as a transferable analytics framework that other EMS services can reuse (National Highway Traffic Safety Administration, 2025; Defence Research and Development Canada, 2017).

Changes to Proposal

Since submitting the initial proposal, the project has evolved in several important ways to better match the realities of the NEMSIS dataset, technical constraints, and the 8-week timeline. The scope has narrowed from broad demand and injury prediction to a more focused goal of forecasting weekly work-related exposure events, after confirming that the national research data only reliably include a general exposure flag and not detailed injury mechanisms. Data engineering work on an initial January 2024 subset revealed extensive nulls and Not Applicable values in hospital outcome fields, prompting a shift to using the full 2024 exposure-flagged cohort (about 20,000 records) and adopting FACTPCR time as a more stable temporal anchor for weekly aggregation. The modeling strategy has moved from a wide comparison of generic ML algorithms to a time-series-first approach (e.g., SARIMA/Prophet) that better fits the aggregated weekly exposure counts, while the dashboard scope has been streamlined into an exposure-risk forecasting tool rather than a broad multi-injury QA engine. Technically, limitations in the free Databricks environment led to a pivot

toward Microsoft SQL Server and SQL Server Management Studio for heavy preprocessing, with Python used on top for forecasting and exploratory ML. Finally, as stretch work, the project now includes plans to experiment with an additional supervised ML model and a small AI component that generates plain-language “weekly risk briefs” for EMS managers from the forecast outputs.

Change ID	What changed	Details of the change	Justification / reasons
a) Scope	From “forecast demand + predict disabling injuries” to “forecast weekly work related exposure events”	The project now focuses on forecasting weekly counts of EMS work related exposures in 2024 using time series models, instead of jointly modeling demand peaks and multiple injury types.	After contacting NEMSIS TAC about additional provider level injury detail, it was confirmed that, at the national level, the only consistently usable provider safety element for this project is the “suspected EMS work related exposure, injury, or death” flag (eOther.05). The public release dataset does not reliably capture detailed injury mechanisms, and many related elements are Not Applicable/Not Recorded. Narrowing to exposure flagged events aligns with what the national data and 8 week timeline can realistically support.
b) Outcomes & joins	From assuming rich hospital outcome linkages to using the exposure flag as the primary safety outcome	Initial work attempted to join January 2024 incidents to exposure and hospital outcome fields to identify “disabling” injuries. After filtering to “Yes” in the exposure flag, many outcome variables were either null or coded Not Applicable, especially hospital admission fields.	NEMSIS guidance notes that many elements legitimately use Not Applicable/Not Recorded when not pertinent or not captured. Empirically, these joins showed that hospital outcomes could not support a reliable “disabling injury” definition in the public research dataset. As a result, the exposure flag itself was adopted as the main safety outcome instead of trying to infer severity from sparse outcome fields.
c) Time base	From generic call/incident time across 2019–2024 to using FACTPCR time for 2024 exposure events	The original plan was to use general incident or call timestamps over multiple years. During work with 2024 exposure positive events, some candidate time fields were missing or	FACTPCR is one of the more consistently populated time fields for exposure flagged records and is available as a national element in the research dataset. Using FACTPCR supports a stable weekly

Change ID	What changed	Details of the change	Justification / reasons
		inconsistent after filtering. The project is now using FACTPCR time (patient care report completion time) as the temporal anchor to aggregate exposure events into a weekly series.	aggregation despite gaps in other times, preserves temporal ordering sufficiently for trend/seasonality analysis, and reflects the reality that different NEMSIS time fields vary in completeness.
d) Modeling strategy	From many generic ML models to time series–first (SARIMA/Prophet)	The initial proposal emphasized comparing linear/logistic regression, random forests, and gradient boosting for demand and injury prediction. With the outcome redefined as weekly exposure counts, the modeling plan now centers on SARIMA/SARIMAX and Prophet to forecast 4–12 weeks ahead with prediction intervals.	Once the primary target became an aggregated weekly time series based on the exposure flag and FACTPCR time, the problem was better matched to time series forecasting than to static tabular ML. Time series models are standard for operational forecasting in emergency and acute care settings and provide interpretable trend/seasonality and uncertainty bands. Focusing on these models keeps complexity manageable and improves interpretability for EMS managers.
e) Dashboard scope	From broad QA automation and multi injury hot spots to an exposure risk forecasting dashboard	Originally, the dashboard and automated reports were envisioned to cover demand, multiple injury types, and geospatial “injury hot spots.” The revised scope prioritizes a simpler product: an interactive dashboard showing historical weekly exposure counts, trend/seasonality, forecast ranges, and “original vs updated” forecast views as new data arrive.	Data constraints (limited injury detail, many Not Applicable outcome fields) make a multi injury QA engine less realistic on the national dataset. Concentrating on exposure risk forecasting delivers a coherent, interpretable tool within 80–100 hours: EMS leaders get a clear early warning signal of high risk weeks and can proactively plan staffing, training, and safety protocols. This preserves operational value while staying aligned with what the NEMSIS research dataset and the midterm data engineering work actually support.

Change ID	What changed	Details of the change	Justification / reasons
f) Stretch work: extra ML & AI text briefs	Adding exploratory ML and AI generated summaries as stretch objectives	Beyond the core time series forecasting, the plan now includes (as stretch work) experimenting with at least one supervised ML model (e.g., random forest or gradient boosting) on a smaller feature set to explore whether specific operational patterns (time of day, call type, crew size) are associated with higher exposure levels, plus a prototype AI API that converts forecast outputs into short “weekly risk briefs” for managers.	Recent work shows growing use of ML and language models for clinical analytics, documentation, and summarization, and adding a small exploratory ML model plus an AI generated text layer allows the project to connect with these trends without over committing core scope. Framing these as stretch objectives keeps the main deliverable focused (time series exposure forecasting and dashboard) while still demonstrating awareness of, and initial hands on work with, broader AI capabilities.
g) Agile data strategy	From “small January subset only” to “full 2024 exposure cohort enriched with other elements”	The initial plan was to work very agile on a tiny subset (e.g., January data) and iterate from there until a forecasting app was ready. In practice, the January only work exposed data issues (null joins, Not Applicable hospital outcomes) and was not enough to build a stable weekly series. The strategy now is to use the full set of ~20,000 2024 exposure flagged records as the core dataset for forecasting, and then enrich that core with additional fields from other NEMSIS tables (beyond just FACTPCR time) plus exploratory ML features.	NEMSIS is extremely large (over 60 million 2024 activations in the research dataset, across 43 relational tables), so starting with a small month level slice was a sensible agile step. However, that slice alone did not provide enough stable weekly counts or clean joins for forecasting. Moving to the full 2024 exposure cohort gives enough volume to model weekly patterns, while still keeping the data manageable. Enriching this cohort with selected variables preserves the agile spirit (iterate, test, adjust) but grounds it in a more statistically robust base.
h) Platform / tooling shift	From Databricks Community Edition to SQL Server + SSMS for heavy preprocessing	The initial proposal mentioned using Databricks Community Edition alongside Jupyter for data pipelines. During	Databricks Community Edition is designed for learning and small datasets; users report practical limits around 10 GB of DBFS storage and file quotas, which are

Change ID	What changed	Details of the change	Justification / reasons
		implementation, the combination of DBFS storage limits (around 10 GB and file quotas in CE) and the size of NEMSIS (60M+ activations, 40+ tables) made it impractical to stage and join large slices of the research dataset there. The project has shifted to using Microsoft SQL Server with SQL Server Management Studio (SSMS) to store, join, and preprocess large NEMSIS tables, with Python/Jupyter layered on top for analysis and modeling.	quickly hit when working with multi gigabyte CSVs or many files. By contrast, SQL Server + SSMS is well suited to managing large relational datasets, defining indexes, and running complex joins and aggregations, which matches the structure and scale of the NEMSIS research dataset. This shift keeps the workflow realistic for big data while still allowing use of Python and ML libraries on top of SQL prepared tables.
i) Hypotheses	From seven operational hypotheses (workload, staffing, shift length, fatigue, ergonomics, equipment, culture) to three data-driven exposure-forecasting hypotheses	The project originally defined seven hypotheses about how workload, staffing, shift rules, ergonomics, equipment, and safety culture might influence injury risk. As the data work progressed, it became clear that most of these constructs cannot be measured reliably with the national NEMSIS research dataset. The hypothesis set has therefore been narrowed to three testable hypotheses focused on temporal patterns in exposure counts, the forecastability of weekly exposures with SARIMA/Prophet, and the operational usefulness of these forecasts when presented in an interactive dashboard.	Many of the initial seven hypotheses require detailed staffing, scheduling, equipment, and culture data that are not available or are too sparse/inconsistent in the national elements (e.g., no robust staffing per hour, shift-length histories, or ergonomic-equipment usage fields). Keeping them as formal hypotheses would over-promise what the dataset can support. Focusing on three exposure-centred hypotheses keeps the project empirically grounded in fields that are consistently available (exposure flag plus time stamps) while still addressing a meaningful operational question: when exposure risk is likely to rise and how to communicate that risk to EMS managers.

Project Planning and Timeline

Overview

From this point to the end of term, the project is organized around completing an exposure-focused time-series forecasting pipeline, building a SQL-backed data layer, and delivering an interpretable dashboard, with exploratory ML and AI-generated “risk briefs” treated as stretch work. The main changes compared with the original plan are: (1) shifting heavy preprocessing from Databricks Community Edition to SQL Server/SSMS due to free-tier storage limits; (2) focusing on the full 2024 exposure-flagged cohort (~20,000 records) instead of a 200k mixed sample; and (3) prioritizing SARIMA/Prophet forecasting and dashboard integration over a large suite of generic ML models, while still reserving time for a small exploratory ML model and an AI summary component.

Phase Breakdown & Milestones

Phase	Main activities	Weeks (overall)	Key milestones & deliverables	Status
1. Setup, learning & exploratory data analysis	Setup and EDA; Databricks Trial; AI Integration Plan	Weeks 1–2 (plus some Week 3/5 touches)	Jupyter/Python environment configured; NEMSIS sample loaded; initial EDA plots; Databricks Community Edition tested; preliminary AI integration ideas documented.	Completed
2. Data cleaning & pipeline development	Data Cleaning; early Feature Engineering	Weeks 2–3 (with a touch in Week 5)	Basic cleaning of January and early 2024 data; first ETL scripts; initial handling of missing/Not Applicable values; preliminary joins tested.	Completed / Ongoing (core cleaning done, refinement continues for exposure cohort)
3. Exposure-focused data prep	Data Cleaning; Feature Engineering; weekly aggregation	Weeks 3–6	Assemble full 2024 exposure-flagged cohort (~20k rows); finalize time field (e.g., FACTPCR) for weekly aggregation; implement SQL/ETL to produce weekly exposure counts; basic weekly plots.	Ongoing
4. Time-series model	Model Development	Weeks 3–7	Build baselines and SARIMA/Prophet on weekly	Ongoing

Phase	Main activities	Weeks (overall)	Key milestones & deliverables	Status
development & selection	(Simple & Advanced); Model Selection		exposure counts; compare models; choose final forecasting approach; document performance and rationale.	
5. Dashboard development	Dashboard Development & AI	Weeks 4, 7–9	Streamlit app that: lets users choose SARIMA or Prophet, select forecast horizon via slider, and optionally include scenario flags (e.g., concerts, political rallies); displays historical data with forecasted weeks; shows a table of forecasts with lower/upper bounds and predicted exposures; includes a second plot comparing original vs updated forecasts as new data are added; and ends with a short, auto-generated recommendation for EMS managers.	Ongoing (core app skeleton and interactions implemented; further polishing and integration with final models still needed)
6. AI-driven reporting (stretch)	AI-Driven Reporting; Dashboard Development & AI	Weeks 7–8	Refine the “recommendation” section of the app into a more systematic AI-generated “weekly risk brief” based on forecast outputs and scenario flags; integrate into the dashboard layout.	Ongoing / early prototype (basic recommendation section exists; AI API integration still to be finalized)
7. Final polish & buffer	Final Polish / Buffer	Weeks 4, 8–9	Clean and comment SQL and Python code; finalize figures and dashboard UI; ensure original vs updated forecast comparison works with new data; complete report and presentation.	Not started / some Week 4 buffer already used

Gantt Chart

Activities	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9
Setup and EDA									
Databricks Trial									
AI Integration Plan									
Data Cleaning (incl. joins, exposure cohort)									
Feature Engineering / Weekly Aggregation									
Model Development (Time-series: baseline + SARIMA/Prophet)									
(Advanced ML) – de-scoped									
Model Selection / Signal Assessment									
AI-Driven Reporting (text recommendations)									
Dashboard Development & AI (Streamlit app)									
Final Polish / Buffer									

Implemented Features

Implemented feature	Description (what it does)	Design & implementation details	Contribution
1. Exposure-focused data pipeline	Builds a 2024 EMS exposure cohort (~20k records) and prepares it for weekly forecasting.	Imported NEMSIS 2024 tables into SQL Server; created SQL queries/views to filter work-related exposure records, handle Not Applicable/Not Recorded codes, and join FACTPCR and related time fields for each record; exposed cleaned views to Python for analysis.	Designed the schema for the SQL tables/views, wrote the SQL joins and filters, tested joins on January data, and iteratively refined them to cover all 2024 exposure records.
2. Weekly exposure time series	Produces a validated weekly time series of	In Python (Pandas), connected to the SQL view, converted FACTPCR timestamps to dates,	Implemented the aggregation code, created initial EDA plots,

Implemented feature	Description (what it does)	Design & implementation details	Contribution
	exposure counts for 2024.	aggregated counts by ISO week, and generated QC plots (line charts, summary stats) to verify patterns and detect anomalies or missing weeks.	and checked consistency between SQL counts and Python results, adjusting filters when weeks did not sum correctly.
3. Time-series forecasting core (SARIMA & Prophet)	Generates short-term forecasts of weekly exposure counts using SARIMA and Prophet.	Built a forecasting module in Python that: splits the weekly series into train/validation; fits simple baselines; then fits SARIMA and Prophet models; returns point forecasts and lower/upper bounds for a user-selected horizon (e.g., 4–12 weeks).	Selected candidate model settings, implemented the training and prediction functions, and wired them so they can be called dynamically by the Streamlit app based on user choices.
4. Interactive Streamlit forecasting UI	Web app where EMS managers can select model and forecast horizon, then view results.	Streamlit layout with sidebar controls: radio buttons to choose SARIMA vs Prophet; slider to choose forecast horizon; optional checkboxes for scenario flags (e.g., concerts, political rallies); “Run forecast” button to execute the chosen model. Main panel displays a combined plot of historical weekly exposures plus forecasted weeks.	Designed the UI flow, created the Streamlit components, and connected them to the forecasting functions so the app recomputes and redraws forecasts when inputs change.
5. Forecast table with intervals	Tabular view of forecast results for managerial review.	After forecasting, the app displays a table listing each forecast week, point estimate, and lower/upper bounds for exposure counts. Data are derived from the SARIMA/Prophet output and formatted for readability.	Wrote the transformation from raw model output into a tidy DataFrame and implemented the Streamlit table display (including column naming and basic formatting).
6. Original vs updated forecast comparison	Visualizes how forecasts change as new weeks of data are added.	Implemented logic to store or simulate an “original” forecast (based on data up to an earlier cut-off) and compare it to an “updated” forecast that includes newer weeks; the app plots both trajectories so managers can see forecast stability.	Designed the comparison concept, coded the function that rebuilds forecasts using different data cut-offs, and created the second plot that overlays original vs updated curves.

Implemented feature	Description (what it does)	Design & implementation details	Contribution
7. Manager recommendation section (AI-driven reporting prototype)	Generates a short narrative recommendation at the bottom of the dashboard.	Simple rules (and planned AI API integration) convert forecast patterns and scenario flags into 1–2 paragraph “risk briefs” (e.g., highlighting weeks with elevated exposure risk and suggesting staffing/safety actions).	Drafted the initial template text, implemented the current rule-based summary, and prepared the hooks for future AI-API integration to upgrade this into an automated text-generation component

AI Use Section

AI Tool Name	Version Account Type	Specific Feature for which the AI Tool was Used	Value Addition (What value did you add over and above what AI did for you?)
Perplexity	Education Pro	Asked AI to surface literature that would support framing the problem that demonstrates occupational hazard encountered EMS personnel	Having done a coop role in VCH, I learned that work-related injuries are handled by WorkSafeBC so I also asked AI to check for any statistics coming from the organization, this is on top of the 2 it produced on CUPE and JEMS
Perplexity	Education Pro	Asked AI to confirm if there’s any AI or ML related studies or projects published using the NEMSIS data in order to check uniqueness of my study	Gave input to AI about the focus of my studies which is really about forecasting work-related injuries for emergency personnels so that it serves as comparison for other studies that may have been done
Perplexity	Education Pro	Asked AI to evaluate my timeline if reasonable and for any crucial activities I may have missed	Provide the milestones I crafted as well as the activities envisioned so that AI provide further input on anything I missed
Perplexity	Education Pro	Asked AI to help build and code the Streamlit EMS forecasting app.	Provided the requirements from selecting models like SARIMAR or prophet, week duration, special features like taking into account

			concerts and more as well as driving how the main content look like with bargraphs and recommendation.
Perplexity	Education Pro	Asked AI to review SQL codes particularly in troubleshooting the time format of the hospital admit time.	Discovered that with my SQL codes, no data is being filtered out so investigated why and found that time format is somehow messing up the filter so I asked AI 3 scenarios on the format.
Perplexity	Education Pro	Asked AI to refine the video script I created, improve the flow and make it more concise so that it's at least 6 minutes long.	Created the outline and initial video script, discussing the major contents I would like to be present in the video then feed it to AI for further improvement.
Perplexity	Education Pro	Asked AI to provide a crash tutorial on Streamlit – app to use and how it works	Provided requirements to build an app and to have dashboard. AI is able to narrow down to Streamlit as a platform that's able to meet my requirements.
Perplexity	Education Pro	Asked AI to provide options for apps that can process large dataset aside from databrick, given that the app is only able to handle 5-10gb data.	Provided the context that I would like to go with Databricks but given its limitations, I would like to explore more app.
Perplexity	Education Pro	Asked AI to layout a couple of approaches for me in terms of joining large dataset in Jupyter Notebook.	Provided context that I'm able to use SQL to narrow down first dataset but given I'm joining to other large dataset, Jupyter Notebook code kept on crashing.

Work Log

Week	Date	Number of Hours	Description of Work Done
1	18-Jan-26	0.5	Meeting with Priya, initial discussion of topics
	21-Jan-26	2	NEMSIS Data initial proposal
	23-Jan-26	0.5	Meeting with Solaris, Tony Tsui

Week	Date	Number of Hours	Description of Work Done
	24-Jan-26	1	Initial works on the final proposal, intro and proposed research project
2	25-Jan-26	2	Align Intro and Proposed Research Project on the final proposal – Riipen link finally given
	26-Jan-26	4	Final works on the final proposal – Estimate and update timelines, build contract, update references, update AI usage table and prompts and perform final review and revisions
	30-Jan-26	2	Review of NEMSIS website, submission of data request for the project, communication of updates to Tony Take Datacamp course on Introduction to Databricks, Setup a community edition Databricks account,
	31-Jan-26	2.5	Review data sent by NEMSIS, extract data (about 2 hrs. processing to extract 200GB data), build initial python code to review data
3	01-Feb-26	2	Update project plan to Agile format, reconsider performing a subset of data performing activities of the whole project in a shorter span of time, create ems eda notebook
	03-Feb-26	2.5	Review data dictionary form NEMSIS and study the headers in the raw dataset. Align equivalent mapping in the data dictionary to get more meaning.
	05-Feb-26	2.5	Provide updates to Riipen stakeholder, further revise project plan. Update ems eda notebook
	06-Feb-26	1.5	Provide updates to Priya, consolidate and review all generated files including notebook, upload to github
	07-Feb-26	1.5	Build end-to-end notebook pipeline from EDA to initial model application and evaluation
4	08-Feb-26	2.5	Update notebook pipeline, troubleshoot time data format, review next steps for the succeeding week.
	10-Feb-26	2	Filter Jan2024 from PCR_Events table using MS SQL SM.
	12-Feb-26	2.5	Analyze Exposure data. Join with PCR Events – results: Null Values for eOther for Jan2024.
	14-Feb-26	3	Review count of YES/No records for Exposure data. Check hospital admit time of Yes data – results: Not Applicable.
5	15-Feb-26	2.5	Revisit hypotheses and pivot. Analyze FACTPCR dataset. Focus on 20K records, eg weekly exposure forecast using SARIMA or Prophet.
	16-Feb-26	2	Design dashboard for forecasting app. Ask forecaster friends some features and functions that are ideal for forecasting app.
	17-Feb-26	2.5	Initiate and code Streamlit app. Include sidebar settings and main context for dashboard.

Week	Date	Number of Hours	Description of Work Done
	18-Feb-26	2	Reach out to NEMSIS report for additional provided injury data –got response that they don’t have the data readily available. Build script for video. Finalize Streamlit app.
	19-Feb-26	3	Shoot and edit video
	20-Feb-26	3.5	Finalize video – add in slides, Initiate midterm report, update readme file
	21-Feb-26	2	Review and polish midterm checklists. Update AI Usage table and List of Prompts table. Upload video in youtube.
6	22-Feb-26	1	Finalize and upload midterm report to Blackboard

Closing and References

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Journals and Articles

Alhazmi, F., Alshamrani, A., Nazer, R., Alshehri, A., & Almutairi, M. (2024). Prevalence of work-related musculoskeletal disorders and associated factors among emergency medical services personnel in Jordan. *BMC Musculoskeletal Disorders*, 25, Article 344.

Hammad, M., Al Nawaiseh, N., Al-Hourani, H., & Al-Dwairi, A. (2025). Prevalence and associated factors of work-related musculoskeletal disorders among emergency medical service workers. *Scientific Reports*, 15, Article 12345.

Marsh, E., Xia, T., & Collie, A. (2025). Profiling paramedic job tasks, injuries, and physical fitness. *Applied Ergonomics*, 114, 104153.

Maguire, B. J., O’Neill, B. J., & O’Meara, P. (2023). Occupational injuries and illnesses among paramedicine clinicians in the United States. *Journal of Occupational and Environmental Medicine*, 65(10), 889–897.

Qian, Z., Rojas, E., & Selevan, J. (2016). Forecasting emergency department hourly occupancy using time series models. *American Journal of Emergency Medicine*, 34(9), 1738–1744.

Reichard, A. A., Marsh, S. M., & Moore, P. H. (2017). Occupational injuries and exposures among emergency medical services workers. *Prehospital Emergency Care*, 21(4), 420–431.

Tarisyai, K., Silva, D., & Ferreira, J. (2024). Enhanced forecasting of emergency department patient arrivals using SARIMAX and machine learning approaches. *Scientific Reports*, 14, Article 23456.

Shahidian, N., et al. (2025). Short-term forecasting of emergency medical services demand using machine learning techniques. *Omega*.

Vieira, S., Cardoso, T., & Ferreira, J. (2025). Forecasting emergency room patient volumes using XGBoost. *Healthcare Analytics*, 6, 100262.

Vieira, S., Cardoso, T., & Ferreira, J. (2023). Forecasting emergency department visits in the reference hospital of a developing country. *arXiv Preprint arXiv:2602.12808*.

Web Reports / Government and Organizational Reports

Defence Research and Development Canada. (2017). *Canadian paramedic health and wellness project* (DRDC-RDDC-2017-R099).

Edmonton Metro EMS. (2025, June 18). *Disabling injury rate*.

National Emergency Medical Services Information System (NEMSIS). (2019). *eOther.06—Type of work-related injury, death or suspected exposure* (Data Dictionary v3.4.0).

National Emergency Medical Services Information System (NEMSIS). (2024). *2024 NEMSIS public-release research dataset*.

National Emergency Medical Services Information System (NEMSIS). (2025). *National EMS Database: NEMSIS research data set user manual* (RDS 350).

National Emergency Medical Services Information System (NEMSIS). (2025). *National EMS database: 2024 data report*.

National Highway Traffic Safety Administration. (2025). *Nationwide EMS incident data*.

WorkSafeBC. (2001). *Evaluation of paramedics' tasks and equipment to control the risk of musculoskeletal injury* (Finding Solutions Grant 99FS-14).

Theses / Dissertations

Jones, C. (2024). *Machine learning-based demand prediction in Dublin, Ireland* (Master's thesis, National College of Ireland).

Appendix

List of AI Prompts

Prompts
Given my project and objective on analyzing EMS data and forecasting information to improve QA protocols to lessen work-related injuries of emergency personnel, find relevant literature that would support demonstrating the occupational hazards encountered by these workers.
When the literature was given, I further prompted if AI can give me stats by WorkSafeBC, having learned about the organization during my coop role time in VCH.
Given that I am doing a project on NEMSIS data, provide confirmation on any studies done in the past that is related to AI or ML so that I understand the significance and uniqueness of my study. My study focuses on forecasting work-related injuries for emergency personnels.
Review and evaluate my timeline of the project from EDA, cleaning, model evaluation and selection until dashboard reporting and check if reasonable given the 10-12.5 hrs I can allot weekly for a 2-month period. Mention any activities I may have missed to include.
Help me design and code a Streamlit app for an EMS exposure-risk forecasting dashboard where the user can choose SARIMA or Prophet, select the forecast horizon with a slider, optionally toggle scenario flags (concerts, political rallies), and then see historical plus forecasted weekly exposure counts with prediction intervals.
Review and debug my SQL code for joining NEMSIS tables on PcrKey and converting hospital admit time / PCR completion time fields into a consistent datetime format that I can use for weekly aggregation.
Refine the video script I created for my midterm presentation on EMS exposure forecasting—improve the flow, make it more concise, and keep the runtime around 6 minutes while preserving key technical details.
Provide a crash tutorial on building a Streamlit app for data science projects: how to structure the main script, add sidebar controls, display plots and tables, and run the app locally.
Suggest options for tools or platforms that can process large datasets besides the free Databricks edition, given that it can only handle around 5–10 GB, and explain pros/cons for my NEMSIS project.
Lay out a couple of approaches for joining large relational datasets in Python/Jupyter (e.g., from NEMSIS): when to rely on pandas merges vs doing joins in SQL first, and how to manage memory efficiently.